Result 1 (module family intersection membership). [Topology foundations, Submodule.mem_sInf]
Let K and I be types. Let *field* be a field structure on K. Let M be a module over K. Let *spaces* be a function from I to Subspace(M). Let x be an element of M. Then

$$x \text{ belongs to } \bigcap_i \mathit{spaces}(i) \text{ if and only if } (\text{for every } i : I,\ x \text{ belongs to } \mathit{spaces}(i)) .$$

Adapted from the cited source.

Result 2 (bounded perturbations have certificates). [Bendich, Edelsbrunner, Morozov, and Patel, well-group perturbation family]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume P.interval.lower $\leq$ P.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Assume $X_r(P)$ is a nested pair (`nesting`). Let h be a continuous real-valued map on P.space. Assume h is a perturbation of P bounded by r (`bounded`). Then

$$(\text{there exists } C : \text{CertifiedPerturbation},\ C.\text{data.map} = h) .$$

Adapted from the cited source. Scope: certificate completeness.

Result 3 (certified perturbations are bounded). [Topology foundations, perturbation certificate evidence]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume $X_r(P)$ is a nested pair (`nesting`). Let C be a bounded perturbation with a certified chain square. Then

$$C.\text{data.map is a perturbation of } P \text{ bounded by } r.$$

Result 4 (perturbation image ignores certificate choice). [Topology foundations, proof irrelevance for inclusion certificates]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume $X_r(P)$ is a nested pair (`nesting`). Let C and D be elements of CertifiedPerturbation. Assume C.data.map = D.data.map (`same`). Then

$$\text{image}(C) = \text{image}(D).$$

Result 5 (relative well group membership). [Bendich, Edelsbrunner, Morozov, and Patel, definition of relative well group]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume $X_r(P)$ is a nested pair (`nesting`). Let x be an element of $H(X_r(P))$. Then

$$x \text{ belongs to } U_r(P; H) \text{ if and only if } (\text{for every } C : \text{CertifiedPerturbation},\ x \text{ belongs to } \text{image}(C)) .$$

Adapted from the cited source.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.